
Global CO₂ Emissions: Trends, Regional Shifts, and Per Capita Inequalities

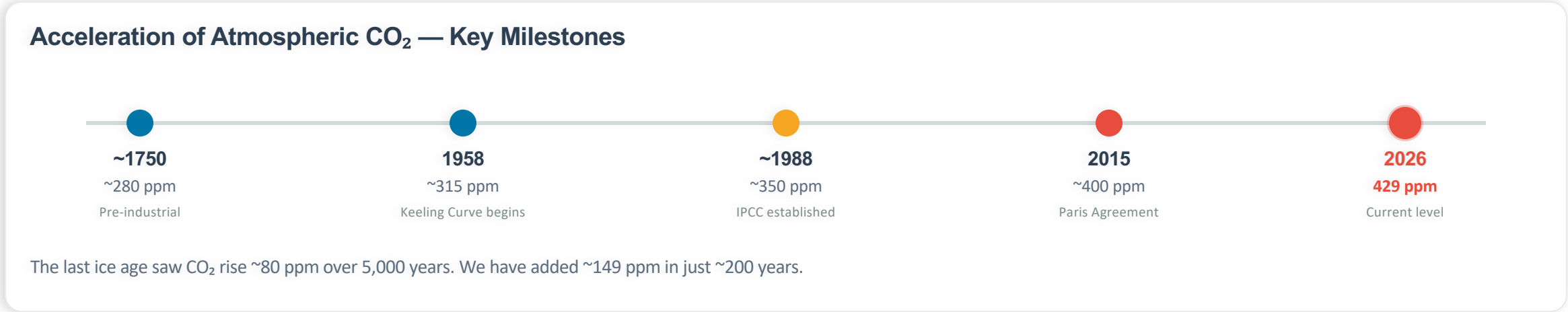
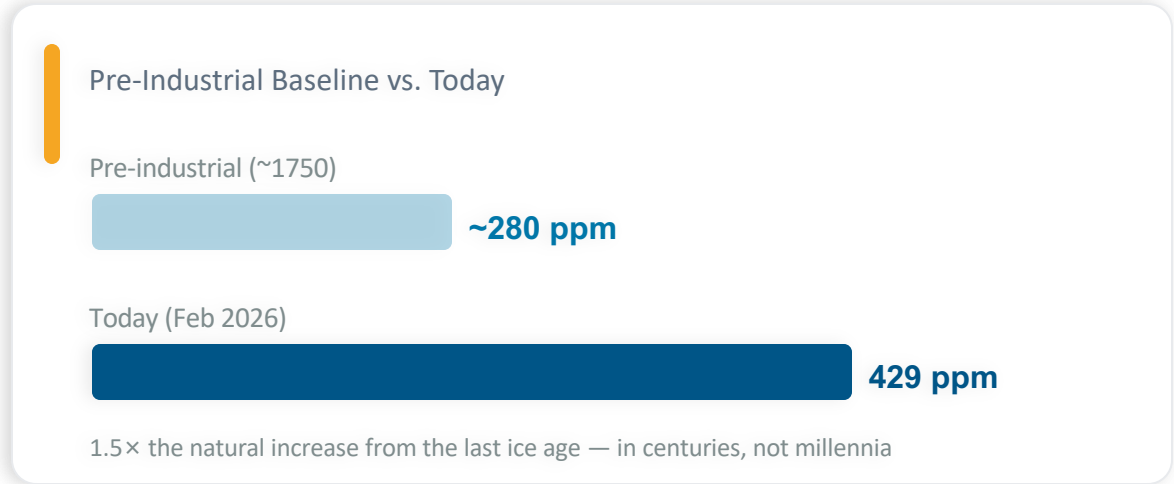
A Data-Driven Overview · 1750–2024

Source: Global Carbon Budget 2025 · NASA · Our World in Data

May 2026

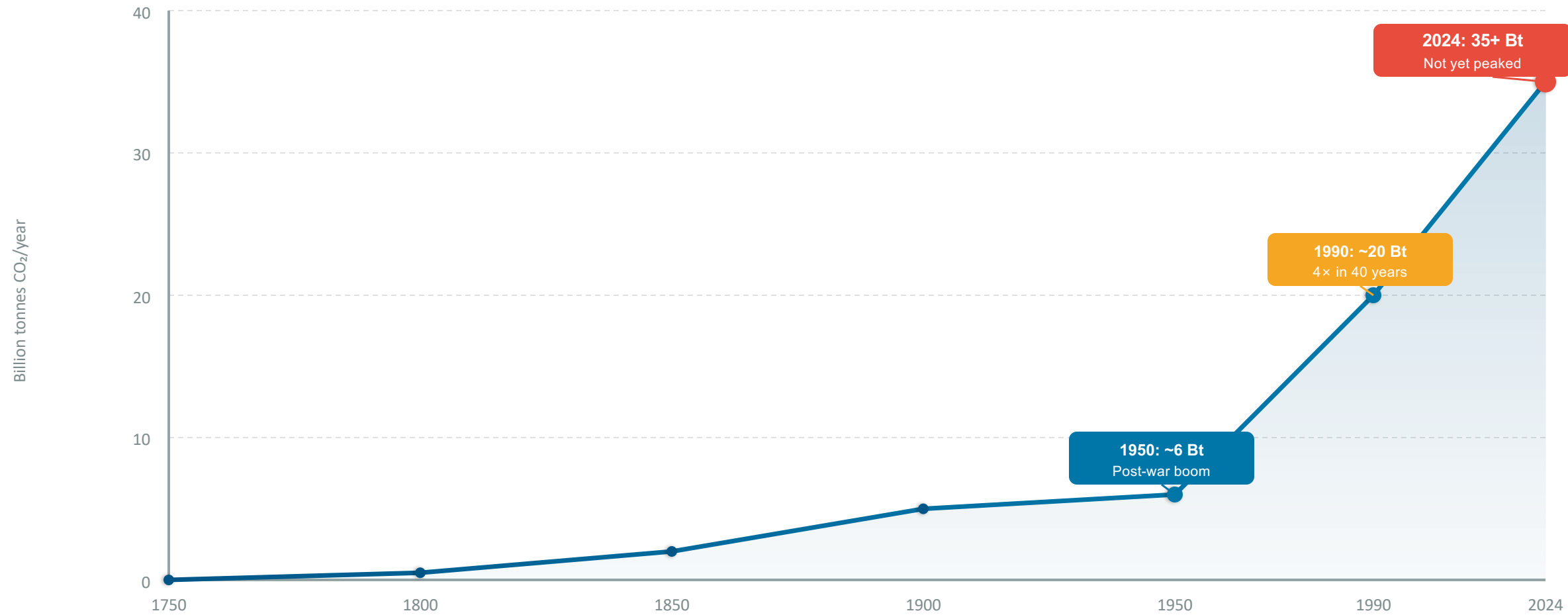
Atmospheric CO₂ Reaches 429 ppm — 50% Above Pre-Industrial Levels

Key insight: CO₂ concentration has risen 53% from the pre-industrial baseline — 1.5× the natural increase from the last ice age



Global Fossil Fuel CO₂ Emissions Grew Six-Fold: 6 Bt to 35+ Bt

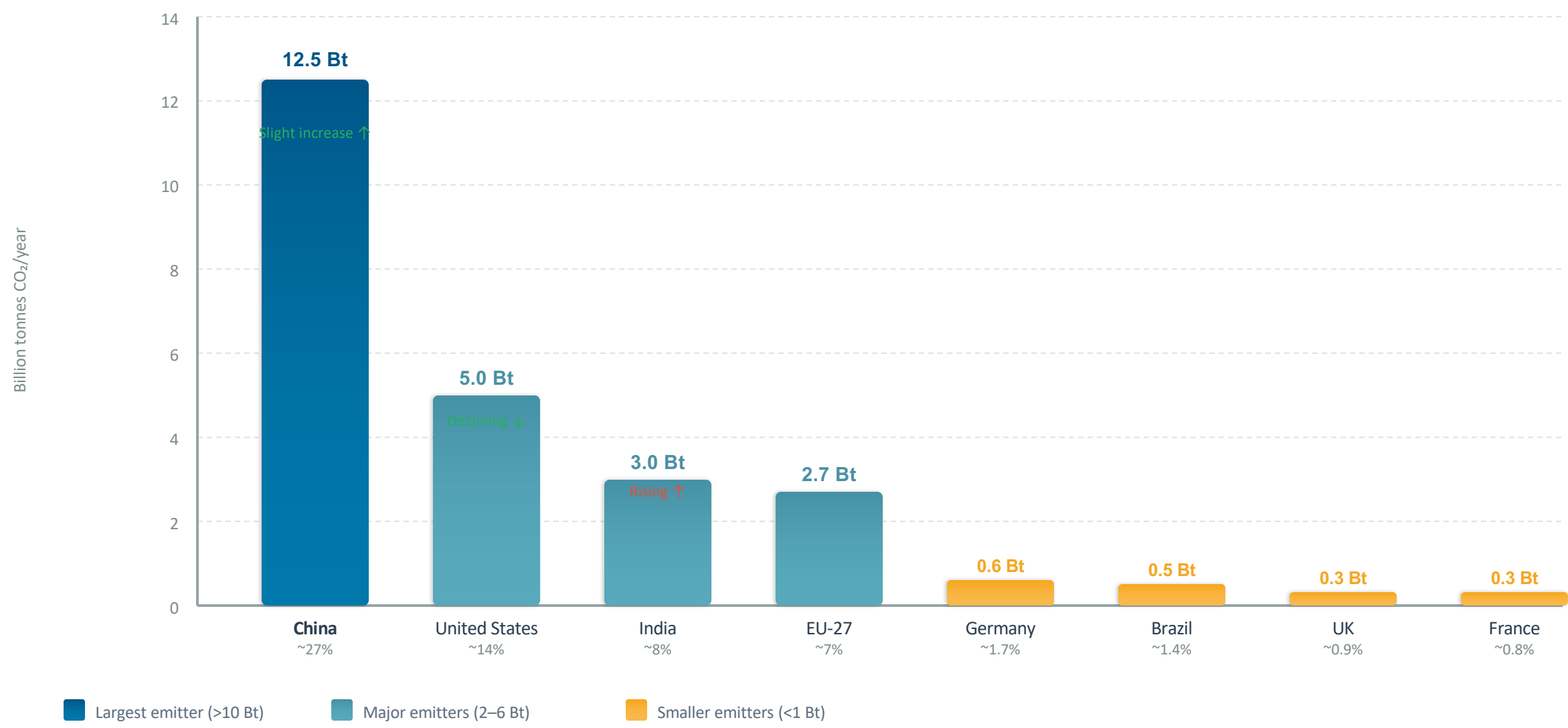
Emissions have grown ~6× since 1950, with growth slowing but not yet peaking



Before the Industrial Revolution, CO₂ emissions were negligible. Growth accelerated sharply after 1950.

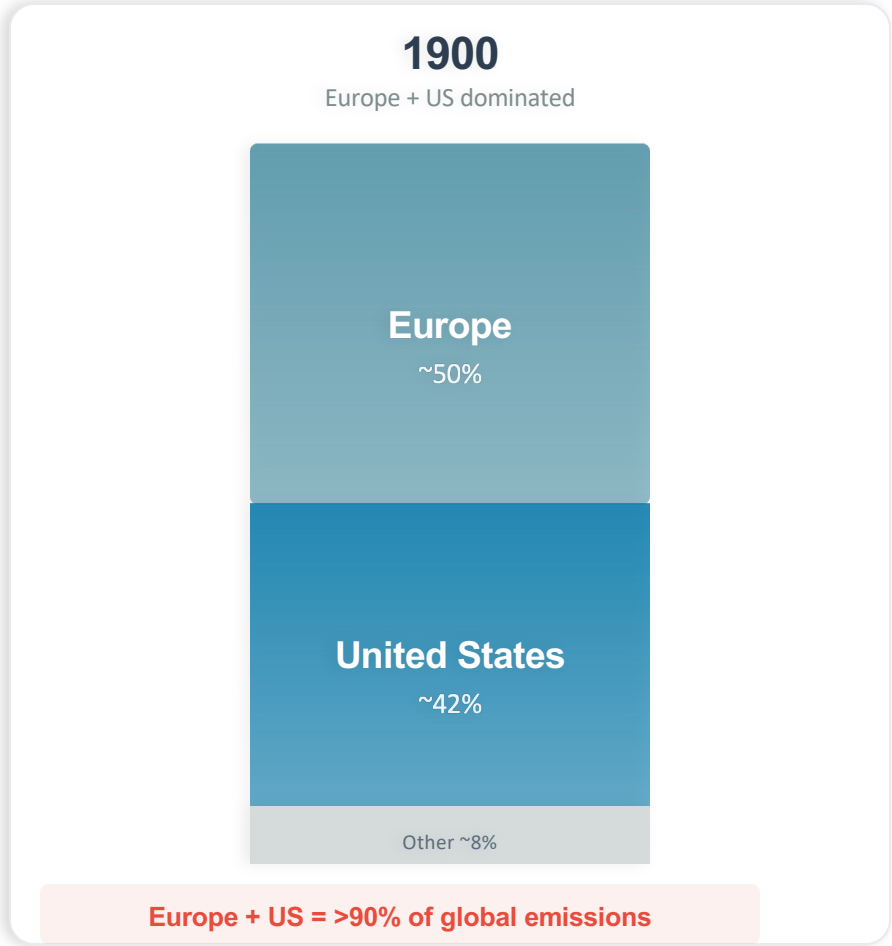
China Now Emits Over 12 Billion Tonnes — More Than Double the US

China accounts for ~27% of global CO₂ — more than the next three emitters combined



Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third

The geographic center of emissions has shifted from >90% in Europe + US (1900) to Asia producing ~50% today



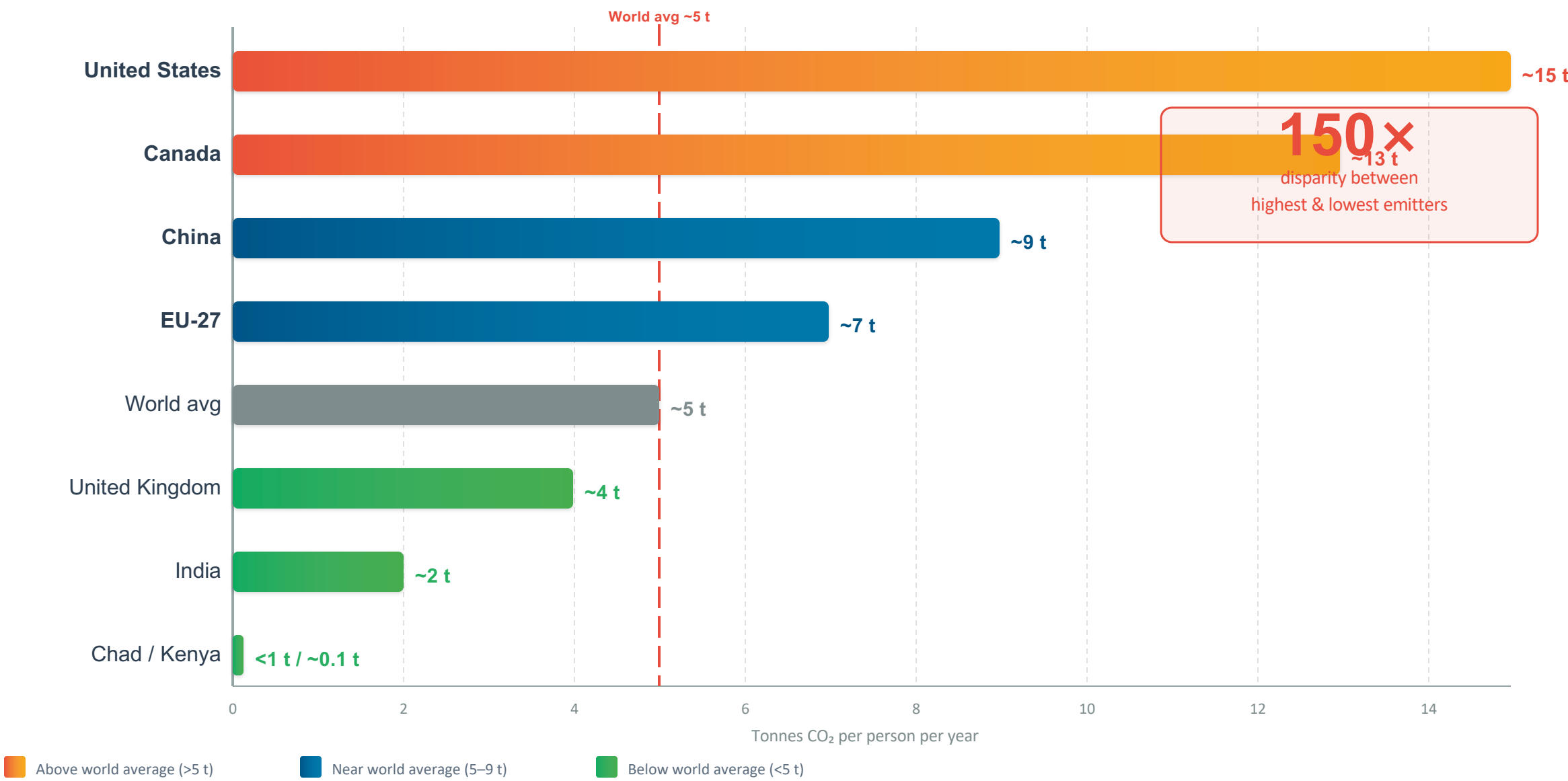
←
124 years



■ Asia ■ United States ■ Europe ■ Africa ■ S. America ■ Other

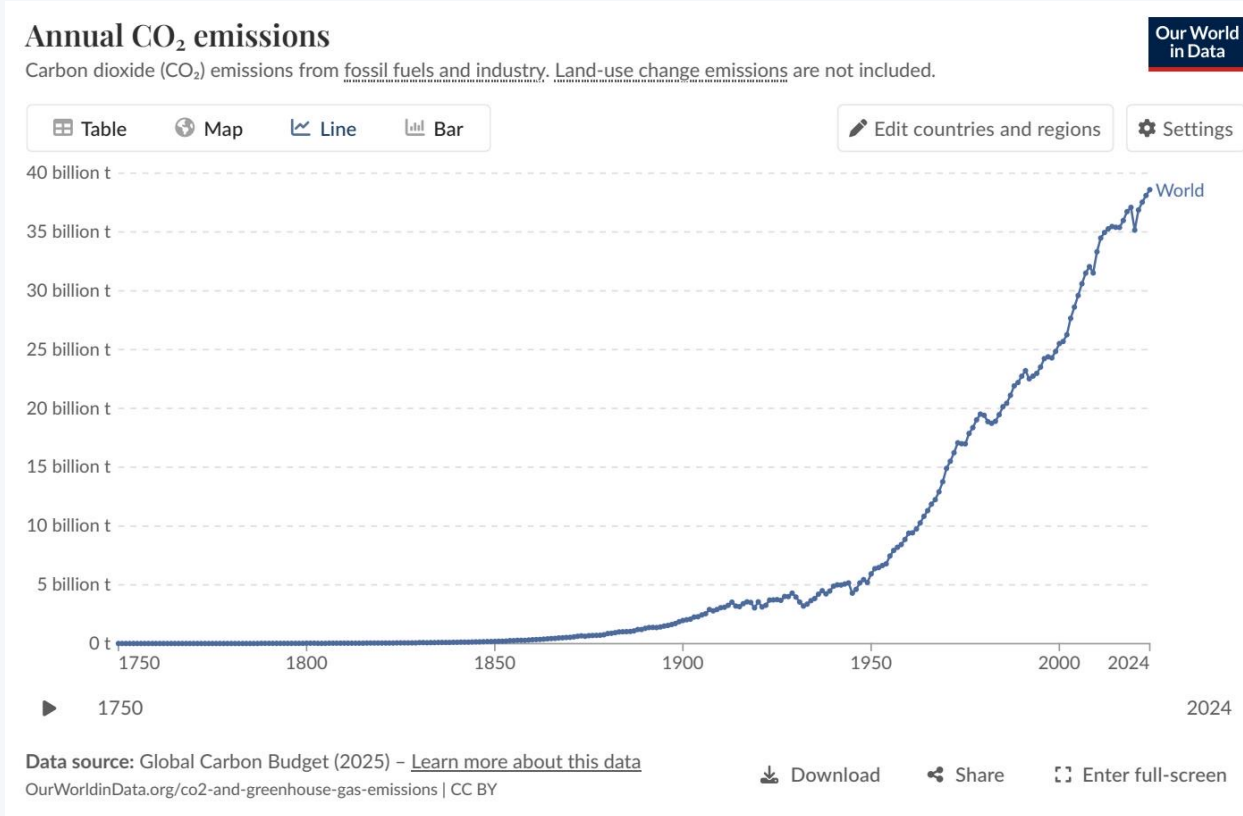
Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity

An American emits in under two days what a person in Chad emits in an entire year



2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

Global emissions map is diverging: developed economies decline while some emerging economies grow rapidly



Declining (–1% to –5%)

United States, much of Europe

Continued structural decline driven by coal phase-out and efficiency gains

Increasing (+2% to +10%)

Russia, parts of Southeast Asia, several African nations

Economic growth outpacing clean energy

! Mixed (varied patterns)

South America: some countries saw very large % increases (>10%) while others saw declines

Map legend: Decline (–1% to –5%) Moderate increase (+2% to +5%) Strong increase (+5% to +10%+)

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Countries with similar living standards can differ significantly in per capita emissions depending on energy sources

Why Does Energy Mix Matter?

Nuclear and renewable energy produce near-zero direct CO₂.

Countries that rely heavily on coal, oil, and natural gas have fundamentally higher per capita emissions — even at similar GDP.



Country	Energy Mix			Per Capita CO ₂
	Nuclear	Renewables	Fossil Fuels	
France	~70%	~12%	~18%	~4 t
UK	~15%	~40%	~45%	~4 t
Germany	~0%	~50%	~50%	~8 t
Netherlands	~3%	~20%	~77% (gas)	~9 t
Takeaway: High nuclear + renewables = lower per capita CO ₂ ; High fossil share = higher per capita emissions				

Key Takeaways

Emissions Are Still Rising, But the Map of Responsibility Is Shifting



Global emissions exceed 35 Bt/year and have not yet peaked

Despite slowing growth rates, the world continues to add CO₂ to the atmosphere at record levels



China alone accounts for >25% of global emissions (~12.5 Bt)

More than double the United States; India is third-largest and rising fast (~3 Bt, ~8%)



Per capita inequalities remain extreme — a 150× gap

US/Australia ~15 t/person vs. Chad/Niger ~0.1 t/person; an American emits in <2 days what a person in Chad emits in a year



US and Europe are declining, but historical cumulative responsibility remains large

From >90% of global emissions in 1900 to <33% today; current declines do not erase cumulative atmospheric burden



Policy and energy mix choices — not just prosperity — determine per capita outcomes

France (~4 t) vs. Germany (~8 t): similar prosperity, 2× difference in per capita CO₂ — driven by nuclear and energy choices